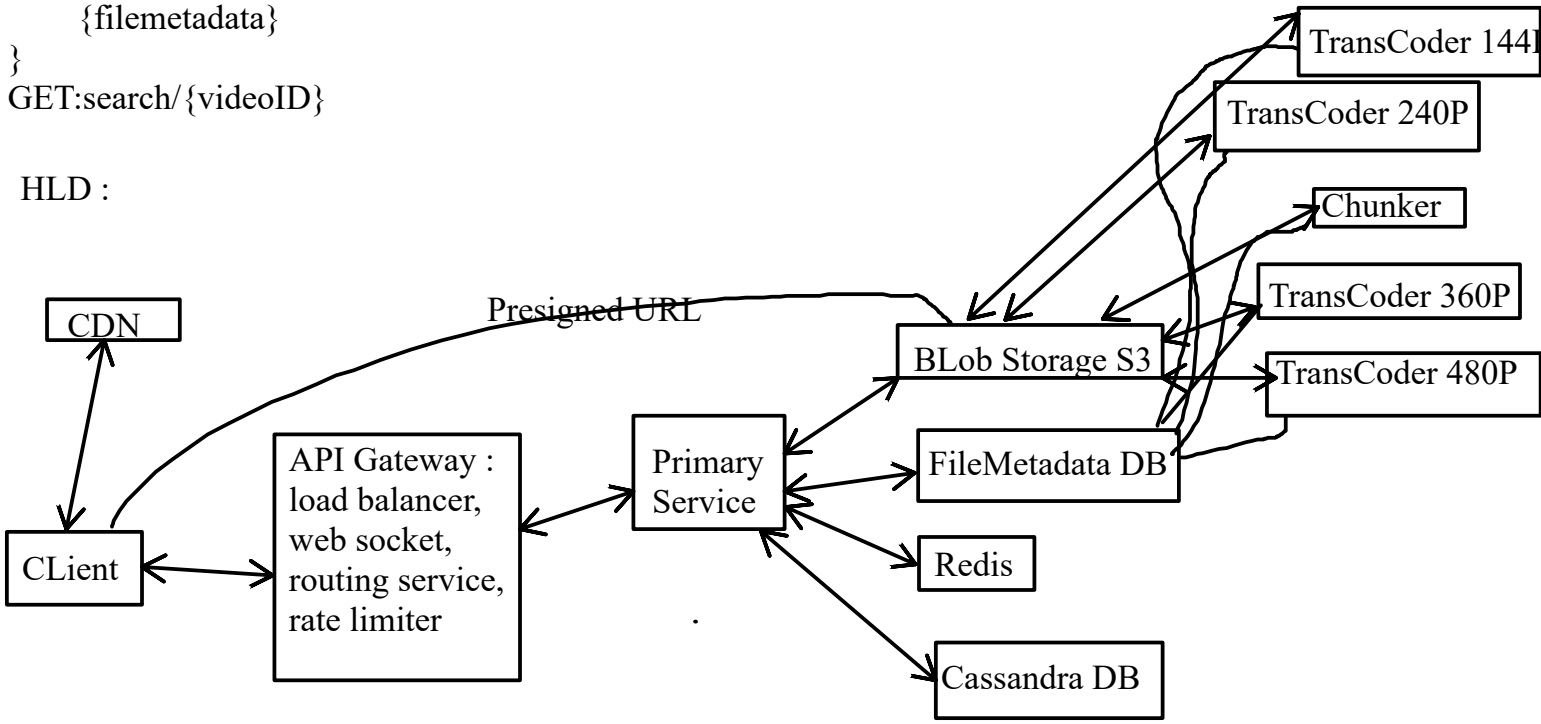
Youtube:

Functional requirements:
User should be able to video any videos based on his preference,
user should be able to adjust videos resolution based on his preference,
user should be able to upload any file>50GB based on his preference.

Non-Functional requirements:
Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime i.e Availability>>Consistency)

Core Entities:
user,
videos,
history,
channel

API :
POST:/upload/video
{
userID,
{filemetadata}
}
GET:search/{videoID}

HLD :